Stopping Is Not Resolving: Why Reasoning Models Loop When No Answer Fits

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Abstract

Reasoning models sometimes never stop thinking, and common fixes simply force them to. We ask why the loops arise and whether stopping them helps. Removing the valid answer from multiple-choice questions makes DeepSeek-R1-Distill-Qwen-7B loop on 73–75% of MATH, GSM8K and AQuA questions it otherwise solves (vs. 2–9%); a Llama-based distill and Qwen3 show the same trigger, and in all three natural AQuA loops concentrate on questions whose open-ended answer matches no option. In R1 and in Qwen3-4B-Thinking, the decision to emit `</think>` is made in late layers in the same way on four benchmarks, but this gate is generic: it is as closed at healthy self-doubt as in loops. Opening it, by forcing `</think>` or with an internal switch learned on controlled loops and tested on held-out natural ones, ends loops without resolving them: in a live test, stopping changes accuracy by -5 to $+30$ points and the switch never beats forcing the token. Resolving the conflict works better where the model accepts it: allowing "None" as an answer removes half or more of the conflict loops in R1 and Qwen3 at almost no cost, but the Llama-based model rarely takes the exit and keeps looping.

1 Introduction

Large reasoning models (LRMs) think before they answer, and occasionally they do not stop: the trace runs to the token budget, re-checking the same step, and no answer is produced (Duan et al., 2026). Practical remedies end the thinking from outside, by forcing the end-of-thinking delimiter (budget forcing; Muennighoff et al., 2025) or by steering the model out of the loop (Yu et al., 2026; Zhang et al., 2026), and are judged on whether the loop is broken. We ask what triggers the loop and whether breaking it resolves whatever caused it.

Our contributions: (1) a controlled trigger: with no valid option, R1 loops on three quarters of multiple-choice questions (MCQs) it can solve, on constructed and real questions, and the same answer–option conflict predicts which natural questions loop (§3); (2) a mechanism: the `</think>` decision is written by late layers, identically across four benchmarks, but the gate is generic rather than suppressed in loops; it is concentrated in a few MLPs in R1 and spread over late MLPs and attention in Qwen3 (§4); (3) stopping is not resolving: in replay and in a live test on held-out loops, every way of stopping ends the thinking but leaves the doubt, and the internal switch never beats budget forcing (§5); (4) resolving can work: allowing "None" removes most conflict loops in two of three models, and fails where the model will not decline (§6).

2 Setup

Models and decoding. DeepSeek-R1-Distill-Qwen-7B (R1; DeepSeek-AI, 2025) and Qwen3-4B-Thinking-2507 (Yang et al., 2025), plus DeepSeek-R1-Distill-Llama-8B (Llama-3.1 base; behaviour only, 2 seeds), served with vLLM (Kwon et al., 2023), sampled with the recommended $T{=}0.6$, top-p 0.95, 4 seeds and a 12k-token budget; for R1 we also report greedy decoding. A **loop** is a run that exhausts the budget without emitting `</think>`.

Conditions. Two-option MCQs built from 150 MATH (Hendrycks et al., 2021) and 200 GSM8K (Cobbe et al., 2021) problems: ALIGNED (correct number vs. an unrelated sentence) and CATCH (a wrong number vs. a sentence: no valid option). To avoid an artificial distractor we also pose 227 AQuA (Ling et al., 2017) and 200 LogiQA (Liu et al., 2020) questions with the correct option deleted, next to their original forms. Natural benchmarks are the datasets of Yu et al. (2026) in native form: GSM8K, MATH-500, AQuA (5 options), LogiQA (4 options).

	valid?	R1	R1 greedy	Qwen3	Llama
MATH-150 ALIGNED	yes	5.3	10.0	9.3	8.0
MATH-150 CATCH	**no**	**74.7**	**98.0**	**29.5**	**71.3**
GSM8K-200 ALIGNED	yes	2.1	8.0	0.0	2.2
GSM8K-200 CATCH	**no**	**73.4**	**100**	0.2	**62.2**
AQuA-227 original	yes	8.5	25.1	2.9	7.7
AQuA-227 correct removed	**no**	**74.4**	**93.8**	**9.0**	**60.1**
LogiQA-200 original	yes	7.5	36.5	2.6	9.0
LogiQA-200 correct removed	**no**	**15.0**	**42.0**	**10.1**	**14.8**
GSM8K / MATH-500 (open)	yes	0.4 / 5.2	4.2 / 10.0	0.1 / 15.3	–
AQuA / LogiQA	yes	7.9 / 9.5	25.6 / 39.4	2.9 / 3.2	–

Table 1: Loop rate (%): the run never emits </think> within 12k tokens. Sampled (T=0.6) rates are means over 4 seeds (Llama: 2). Top: the same questions with and without a valid option; bottom: natural benchmarks.

Learning and held-out testing. For each benchmark and temperature, questions that loop naturally in any seed are *held out*. On the remaining questions we pair a CATCH run that loops with the ALIGNED run that finishes correctly (same question and seed) and compare the token before </think> in the finished run with the loop's *commit point*, where, having just stated an answer, it writes "Wait" (detected from the text without the answer key; hand-checked, Appendix). Stop components and their values (the mean output at real stops) are learned from these pairs only. We test on the held-out questions in two ways. *Replay*: each natural loop is fed up to its commit point and the intervention applied there. *Live*: the held-out questions are generated afresh with new seeds; a rule that reads only the text so far fires on any run with no </think> after 6k thinking tokens, at the next "states a result, then doubts it" point (fixed in advance, not tuned), and every run it fires on, loop or not, is continued under each intervention. Answers (up to 1.2k tokens after </think>) are scored against the key.

3 When do reasoning models loop?

Removing the valid answer makes R1 loop on 73–75% of MATH, GSM8K and AQuA runs, against 2–9% on the same questions with a valid option (Table 1); greedy decoding raises both, to 94–100% without a valid option. The traces are explicit: the model derives the answer, notes it is not an option, re-derives it, and repeats. Runs that do finish pick a wrong option (82% for MATH CATCH, 100% for AQuA) and never decline, as instruction-tuned models do (Góral et al., 2025). LogiQA, whose options are arguments rather than values, only doubles (7.5% to 15%). Qwen3 shows the same effect at a smaller scale (MATH 9% to 30%, AQuA 3% to 9%, LogiQA 3% to 10%) and handles GSM8K CATCH without looping; the Llama-based distill reproduces R1 (MATH 8% to 71%, GSM8K 2% to 62%, AQuA 8% to 60%), so the trigger is not specific to the Qwen base.

Natural loops follow the same conflict. We ask each model the AQuA questions with no options and group them by its own answer. If that answer matches the correct option, the 5-option question loops in 1% of runs (R1; Qwen3 0%, Llama 2%); if it matches no option, through the model's own error or a noisy key, it loops in 30% (Qwen3 10%, Llama 20%). Natural MCQ loops are the controlled conflict, arising by itself.

4 Where is the stop decided?

At a real stop </think> has logit ≈+32 and is the top-1 token; at the loop's commit point it is ≈+5 (gap > 0 in 100% of pairs). Direct logit attribution (nostalgebraist, 2020; Nanda and Bloom, 2022) assigns about 90% of R1's gap to MLPs. Learned separately on each benchmark's own pairs (24–175 pairs per benchmark and temperature), the six MLPs carrying it are the same in all eight settings (layers 22–27), the gap is +24 to +29, and replacing their output at the commit point with the learned values recovers 78–98% of it (control MLP: $\leq$1%).

The gate is generic. If loops suppressed the gate, the stop signal should be lower at a loop's commit point than at a healthy doubt. It is not (Figure 1a): at the same position, loop doubts are +0.35 logits *higher* (95% CI 0.08–0.62), against ≈+26 for a real stop, and across all natural loops the highest P(</think>) at any line break is ≈0.001 (median). The late layers are the model's general "stop now" gate; a loop is a run that never reaches the state that opens it.

Qwen3 spreads the gate. Qwen3 shows the same gap (+27 to +28 on every benchmark), but only 59% of it is written by MLPs and 41% by attention in the last four layers; its six strongest MLPs carry 47%. Patching them recovers half the gap but never makes </think> the top token. Patching all MLPs from layer 18 recovers 83% and makes it the top token in 93% of pairs, and replacing the residual stream at layer 21 or later recovers all of it: the decision is formed by layer 21 and written by many late components.

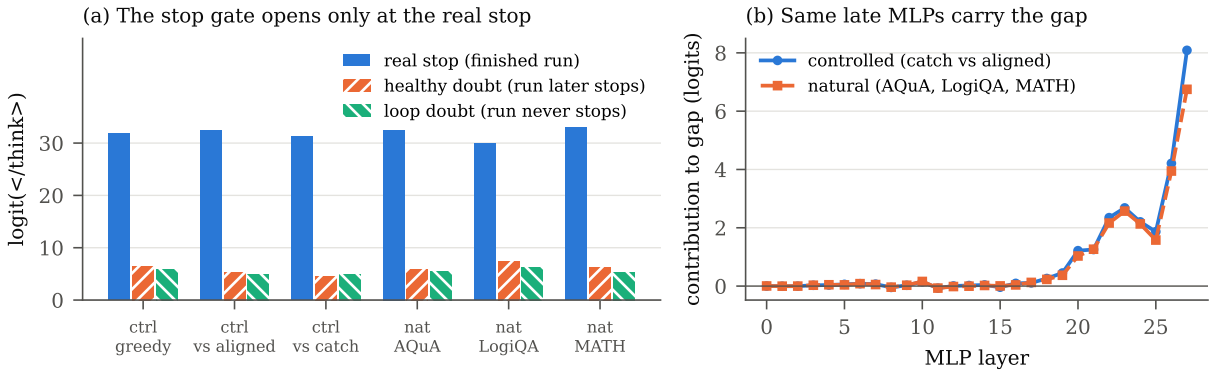


Figure 1: R1. (a) Logit of `</think>` at the real stop of finished runs, at healthy doubts in finished runs (an answer is stated, then "Wait", but the run later stops) and at the commit point of looping runs. (b) Direct contribution of each MLP to the gap between the real stop and the loop's commit point, for controlled and natural pairs.

	R1			R1 greedy			Qwen3		
	none	force	switch	none	force	switch	none	force	switch
AQuA	31	39	41	28	58	52	38	46	38
LogiQA	31	37	34	22	37	33	42	41	41
MATH-500	45	40	40	42	44	46	34	60	34
GSM8K	–	–	–	38	57	57	–	–	–

Table 2: Live test: accuracy (%) over fresh runs of the held-out questions (those that loop naturally) when the stop rule may fire. none = no intervention; force = `</think>`; switch = the benchmark's own learned stop MLPs, set once. 95% CIs are ±8–13 points. Sampled GSM8K has too few held-out questions (R1: 5, Qwen3: 1).

	R1		Qwen3		Llama	
no valid option	loops	"None"	loops	"None"	loops	"None"
MATH CATCH	76 → **35**	44	32 → **18**	82	71 → 67	8
GSM8K CATCH	72 → **19**	66	0 → 0	96	62 → 60	17
AQuA, correct removed	74 → **35**	30	9 → **4**	75	60 → 51	14
LogiQA, correct removed	16 → **8**	20	10 → 8	18	15 → 10	11

Table 3: Allowing "None": loop rate (%) without → with the added sentence, and the share of runs answering "None" (2 seeds each).

5 Does stopping fix loops?

Replay. At the commit point of held-out natural loops, forcing `</think>` ends every loop and the learned switch 79–91% of R1's, but answers are right for only 14–36% (R1, T=0.6; switch 15–25%), against 0% if the loop is left alone. The doubt survives the stop: the answer often opens with "But wait" and keeps re-checking until the answer budget runs out (45–86% of answers). In Qwen3 the switch stops 0% of loops (forcing: 17–38% correct), as its gate predicts. A broad patch of all late MLPs does stop all 20 Qwen3 loops we tested, with 10% correct, the same as forcing `</think>`.

Live. As a real stop rule (Table 2), stopping helps on the MCQ benchmarks (R1: +3 to +10 points) and hurts on MATH-500 at T=0.6 (−5): there the rule cuts short runs that would have finished correctly (7) more often than it rescues loops (2). Gains are larger where loops are runs that would have solved the problem: greedy loops, which are mostly repetition (AQuA 28% to 58%), and Qwen3's MATH loops, which mostly run out of budget (34% to 60%). Across all ten settings, the internal switch is never better than forcing `</think>` by more than 2 points, and on Qwen3 it never fires. Opening the gate from inside adds nothing to opening it from outside.

6 Resolving the conflict

If the conflict drives the loop, a legitimate way out should remove it where stopping did not. We add one sentence to the prompt: "If none of the options is correct, put None within \boxed{}." Loops fall by 50–74% for R1 and by 44–56% where Qwen3 loops (MATH, AQuA), and the model now often declines (Table 3). The cost is small: on the same questions with a valid option, the models answer "None" in 0–5% of runs and accuracy changes by at most 2 points (Llama LogiQA: 5). The exit only helps if it is taken: the Llama-based distill answers "None" in just 8–17% of runs and keeps looping (MATH 71% to 67%, GSM8K 62% to 60%), and LogiQA runs in every model still pick an option 70–78% of the time. The trigger is shared; the willingness to decline is not.

The exit does not open the gate; reaching a conclusion does. Where the model says its answer is not among the options, the stop signal is equally low with or without the exit, and whether or not

the run then declines (R1: maximum logit +5.9, +6.1 and +6.3; Qwen3 +0.4 to +1.0; </think> is never the top token there; 60 runs per group). It rises only at the run's actual stop, after it has written "None" (+30 in R1, +26 in Qwen3), as at any real stop. The exit works upstream of the gate: it gives the reasoning a conclusion it can reach, and the generic gate then opens as usual.

7 Related work

Loops in LRMs have been characterised through attention (Duan et al., 2026; Zhang et al., 2026) and broken by steering (Yu et al., 2026) or budget forcing (Muennighoff et al., 2025); we isolate a trigger and show that breaking a loop does not resolve it, while removing the trigger does. Missing premises cause verbose reasoning (Fan et al., 2025), and instruction-tuned models pick invalid options (Góral et al., 2025); a reasoning model facing the same conflict fails to terminate. Hidden states encode answer correctness (Zhang et al., 2025); acting on this signal with a forced stop does not recover it (Appendix B).

8 Conclusion

Reasoning models loop when their answer cannot be reconciled with the options, a condition we create on constructed and real questions and that predicts which natural questions loop. A generic late-layer gate decides when to stop, and loops never open it. Forcing it open, from outside or inside, ends the thinking but not the doubt; giving the model a legitimate answer for the conflict removes most loops, when the model is willing to take it. Loop-breaking methods should be judged on what happens after the stop, and resolving loops requires acting on their cause.

Limitations

We study three models of 4–8B parameters with a 12k-token budget (mechanism on two); larger or differently trained LRMs may differ, and Qwen3 is evaluated only with sampling, as its authors recommend. Commit points come from a text heuristic that does not use the answer key (hand-checked: 11/12 controlled, 9/10 AQuA/LogiQA; on MATH about 2 in 3 land on a stated result). Held-out test sets are small for some settings (sampled GSM8K 5 and 1 questions; Qwen3 AQuA 13), and LogiQA yields few learning pairs (24–26). The live test's runs up to the trigger are generated with vLLM and continued with TransformerLens; its trigger threshold was fixed in advance and not tuned. Activations are bfloat16, to which late layers are sensitive. We do not identify the upstream computation that opens the gate.

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A Implementation details

Mechanism analyses use TransformerLens with RMSNorm folding and unembedding centering applied in place on the GPU (matching an unprocessed model to bf16 precision), KV-cached batched decoding, and positions mapped through tokenizer character offsets. A finished run is rebuilt as reasoning + `</think>` + answer exactly as generated: inserting an extra newline before `</think>` moves the "stop" to a position where $P(\texttt{</think>}) \approx 0$. Commit points: the first sentence-initial doubt ("Wait", "But", "Alternatively", ...) after a sentence that states the value the loop states most often (MCQs: with a conclusion cue such as "answer" or "not among the options"; open-ended: a sentence ending in a stated result, excluding conditions and checks). An earlier version that preferred the gold value for open-ended loops inflated MATH replay accuracy (38% vs. 14% with forcing) and was replaced. MATH answers are scored with math-verify. Code and data will be released.

B The model knows its answer is right, but a stop does not use it

A logistic-regression probe on R1's layer-24 residual at 6.1k "states an answer, then doubts it" points from 1.6k sampled runs, labelled by the answer key with 5-fold splits by question, predicts whether the stated answer is correct with AUC 0.88 overall and 0.77 at loop points (MATH-500 0.87, AQuA 0.80, LogiQA 0.53). Used as an online stopping rule on recorded loops (stop at the first point with held-out $P(\text{correct}) > 0.7$), it picks better stated answers (40% vs. 21% correct) but not better written answers (19–36%), and its overall accuracy (8–18%) is below stopping every loop (14–28%).